

Universe and Space

MCQ

1. The Universe includes

- a) Only planets
- b) Only stars
- c) Everything that exists
- d) Only living things

Answer: c

2. Before the birth of the Universe, there was no

- a) Gravity
- b) Light
- c) Time, space and matter
- d) Energy

Answer: c

3. The study of the Universe is called

- a) Biology
- b) Geography
- c) Astronomy
- d) Geology

Answer: c

4. The diameter of the known Universe is about

- a) 9.3 billion light-years
- b) 93 billion light-years
- c) 930 million light-years
- d) 9.3 trillion light-years

Answer: b

5. Astronomical distances are measured in

- a) Kilometres
- b) Meters
- c) Light-years
- d) Seconds

Answer: c

6. One light-year is the distance travelled by

- a) Sound in a year
- b) Light in a second
- c) Light in a year
- d) Earth in a year

Answer: c

7. 1 light-year is approximately

- a) 9.46×10^6 km
- b) 9.46×10^9 km
- c) 9.46×10^{12} km
- d) 9.46×10^{15} km

Answer: c

8. The Universe is made up of

- a) Only galaxies
- b) Only stars
- c) Matter and energy
- d) Only planets

Answer: c

9. A galaxy is a collection of

- a) Planets only

b) Stars and celestial bodies

c) Satellites only

d) Comets only

Answer: b

10.The Sun and objects revolving around it form

a) Galaxy

b) Constellation

c) Solar system

d) Universe

Answer: c

11.Our solar system is located in

a) Andromeda

b) Orion

c) Milky Way

d) Ursa Major

Answer: c

12.A planet is a body that

a) Emits its own light

b) Orbits a star

c) Orbits a satellite

d) Twinkles

Answer: b

13.A satellite is an object that

a) Emits light

b) Orbits a larger body

c) Is stationary

d) Is a star

Answer: b

14.The Universe contains

a) One galaxy

b) Few galaxies

c) Billions of galaxies

d) Only Milky Way

Answer: c

15.The word “Universe” comes from

a) Greek

b) Sanskrit

c) Latin

d) Arabic

Answer: c

16.Geocentric model states that

a) Sun is at the centre

b) Earth is at the centre

c) Moon is at the centre

d) Galaxy is at the centre

Answer: b

17.The geocentric model was supported by

a) Galileo

b) Newton

c) Aristotle

d) Copernicus

Answer: c

18.Ptolemy lived in

- a) 6th century BC
- b) 2nd century AD
- c) 10th century AD
- d) 17th century AD

Answer: b

19.Ancestors believed Earth is stationary because

- a) It rotates
- b) They felt no motion
- c) Sun is stationary
- d) Moon moves fast

Answer: b

20.Apparent daily motion of celestial objects is from

- a) West to east
- b) North to south
- c) East to west
- d) South to north

Answer: c

21.The Moon completes one revolution around Earth in about

- a) 15 days
- b) 27 days
- c) 29.5 days
- d) 365 days

Answer: b

22.Moon phases repeat once every

- a) 15 days

- b) 27 days
- c) 29.5 days
- d) 30 days

Answer: c

23. On a New Moon day, the Moon is

- a) Half visible
- b) Fully visible
- c) Not visible
- d) Crescent shaped

Answer: c

24. A fully illuminated Moon is called

- a) New Moon
- b) Half Moon
- c) Full Moon
- d) Crescent Moon

Answer: c

25. Epicycle is a

- a) Straight path
- b) Elliptical path
- c) Small circular orbit
- d) Zigzag path

Answer: c

26. Epicycles were used to explain

- a) Gravity
- b) Seasons
- c) Planetary motion

d) Eclipse

Answer: c

27.The larger orbit in epicycle model is called

a) Orbit

b) Deferent

c) Axis

d) Arc

Answer: b

28.Retrograde motion refers to

a) Forward motion

b) Circular motion

c) Backward motion of planets

d) Stationary motion

Answer: c

29.Retrograde motion is an

a) Actual motion

b) Apparent motion

c) Random motion

d) Linear motion

Answer: b

30.Telescope was used by

a) Newton

b) Copernicus

c) Galileo

d) Kepler

Answer: c

31. Telescope was invented by

- a) Galileo
- b) Newton
- c) Hans Lippershey
- d) Copernicus

Answer: c

32. Galileo observed mountains on

- a) Earth
- b) Mars
- c) Moon
- d) Venus

Answer: c

33. Galileo discovered moons of

- a) Mars
- b) Jupiter
- c) Saturn
- d) Venus

Answer: b

34. Rings were observed around

- a) Jupiter
- b) Mars
- c) Saturn
- d) Uranus

Answer: c

35. Phases of Venus supported

- a) Geocentric model

- b) Epicycle model
- c) Heliocentric model
- d) Big Bang theory

Answer: c

36. Heliocentric model states that

- a) Earth is centre
- b) Sun is centre
- c) Moon is centre
- d) Galaxy is centre

Answer: b

37. Heliocentric theory was proposed by

- a) Galileo
- b) Kepler
- c) Copernicus
- d) Newton

Answer: c

38. Venus always appears near

- a) Earth
- b) Moon
- c) Sun
- d) Mars

Answer: c

39. The origin of the Universe is explained by

- a) Heliocentric theory
- b) Geocentric theory
- c) Big Bang theory

d) Epicycle theory

Answer: c

40.The Big Bang occurred about

- a) 4 billion years ago
- b) 10 billion years ago
- c) 14 billion years ago
- d) 20 billion years ago

Answer: c

41.After Big Bang, the Universe

- a) Contracted
- b) Expanded
- c) Remained same
- d) Disappeared

Answer: b

42.First elements formed were

- a) Oxygen and nitrogen
- b) Hydrogen and helium
- c) Carbon and oxygen
- d) Iron and nickel

Answer: b

43.Galaxies are the

- a) Smallest units
- b) Building blocks of Universe
- c) Satellites
- d) Stars

Answer: b

44. Visible matter forms about

- a) 5%
- b) 27%
- c) 50%
- d) 68%

Answer: a

45. Dark energy forms about

- a) 5%
- b) 27%
- c) 50%
- d) 68%

Answer: d

46. A galaxy is held together by

- a) Magnetism
- b) Nuclear force
- c) Gravitational force
- d) Electric force

Answer: c

47. Average galaxy size is measured in

- a) Kilometres
- b) Light-years
- c) Parsecs
- d) AU

Answer: c

48. Astronomical Unit is distance between

- a) Earth and Moon

- b) Sun and Mars
- c) Earth and Sun
- d) Sun and Jupiter

Answer: c

49.1 AU equals

- a) 1.496×10^8 km
- b) 9.46×10^{12} km
- c) 3.09×10^{13} km
- d) 1×10^6 km

Answer: a

50.A parsec equals about

- a) 1 AU
- b) 1 Ly
- c) 3.26 Ly
- d) 100 Ly

Answer: c

51.Milky Way is a

- a) Elliptical galaxy
- b) Irregular galaxy
- c) Spiral galaxy
- d) Ring galaxy

Answer: c

52.Spiral galaxies contain

- a) Only old stars
- b) Only gas
- c) Bulge and spiral arms

d) No dust

Answer: c

53. Elliptical galaxies contain stars that are

a) Very young

b) Very old

c) Newly formed

d) Exploding

Answer: b

54. Irregular galaxies lack

a) Stars

b) Gas

c) Definite shape

d) Gravity

Answer: c

55. Barred spiral galaxies have

a) No bulge

b) Central bar

c) No arms

d) No stars

Answer: b

56. Our Milky Way is classified as

a) Elliptical

b) Irregular

c) Barred spiral

d) Ring

Answer: c

57.Andromeda is

- a) A star
- b) A constellation
- c) Nearest galaxy
- d) A planet

Answer: c

58.The Milky Way appears as

- a) Dark patch
- b) Bright band
- c) Single star
- d) Circle

Answer: b

59.Solar system is about

- a) 2,700 ly from centre
- b) 27,000 ly from centre
- c) 270,000 ly from centre
- d) 2,700,000 ly from centre

Answer: b

60.Time taken by solar system to orbit Milky Way is about

- a) 23 million years
- b) 230 million years
- c) 2.3 million years
- d) 23 billion years

Answer: b

61.A constellation is

- a) Group of galaxies

- b) Pattern of stars
- c) Group of planets
- d) Real object

Answer: b

62.Total constellations recognized by IAU are

- a) 48
- b) 72
- c) 88
- d) 100

Answer: c

63.Big Dipper is part of

- a) Orion
- b) Ursa Minor
- c) Ursa Major
- d) Andromeda

Answer: c

64.Pole star is located in

- a) Ursa Major
- b) Orion
- c) Ursa Minor
- d) Cassiopeia

Answer: c

65.Constellations change with seasons due to

- a) Rotation of Earth
- b) Revolution of Earth
- c) Rotation of Moon

d) Revolution of Moon

Answer: b

66. Constellations are

a) Bound by gravity

b) Optical patterns

c) Single objects

d) Galaxies

Answer: b

67. Solar system is held together by

a) Magnetism

b) Gravity

c) Nuclear force

d) Friction

Answer: b

68. Number of planets in solar system is

a) 7

b) 8

c) 9

d) 10

Answer: b

69. Pluto is now classified as

a) Planet

b) Star

c) Satellite

d) Dwarf planet

Answer: d

70.Stars are made mostly of

- a) Oxygen and nitrogen
- b) Hydrogen and helium
- c) Carbon and iron
- d) Nitrogen and carbon

Answer: b

71.Stars appear to twinkle due to

- a) Earth rotation
- b) Atmospheric disturbance
- c) Star motion
- d) Moonlight

Answer: b

72.Nearest star to Earth is

- a) Alpha Centauri
- b) Sirius
- c) Sun
- d) Vega

Answer: c

73.Next nearest star after Sun is

- a) Sirius
- b) Alpha Centauri
- c) Proxima
- d) Betelgeuse

Answer: b

74.A satellite revolves around

- a) Star

- b) Galaxy
- c) Planet
- d) Constellation

Answer: c

75. Moon is a

- a) Planet
- b) Star
- c) Natural satellite
- d) Asteroid

Answer: c

76. All planets except _____ have moons

- a) Earth
- b) Mars
- c) Mercury and Venus
- d) Jupiter

Answer: c

77. Artificial satellites are

- a) Natural
- b) Human-made
- c) Stars
- d) Moons

Answer: b

78. First artificial satellite was

- a) Aryabhata
- b) Sputnik
- c) Explorer

d) INSAT

Answer: b

79. Aryabhata satellite belonged to

a) USA

b) Russia

c) India

d) China

Answer: c

80. Artificial satellites help in

a) Weather forecasting

b) Communication

c) Navigation

d) All of these

Answer: d

81. ISRO stands for

a) Indian Satellite Research Organisation

b) Indian Space Research Organisation

c) International Space Research Office

d) Indian Scientific Research Organisation

Answer: b

82. ISRO was formed in

a) 1962

b) 1965

c) 1969

d) 1975

Answer: c

83.ISRO headquarters is in

- a) Chennai
- b) Mumbai
- c) Bengaluru
- d) Delhi

Answer: c

84.Father of Indian space program is

- a) C.V. Raman
- b) Vikram Sarabhai
- c) Homi Bhabha
- d) Abdul Kalam

Answer: b

85.India's first satellite was launched in

- a) 1969
- b) 1975
- c) 1980
- d) 1985

Answer: b

86.Rohini satellite was launched using

- a) PSLV
- b) GSLV
- c) SLV-3
- d) LVM-3

Answer: c

87.Chandrayaan-1 was launched in

- a) 2005

b) 2008

c) 2013

d) 2019

Answer: b

88.Mars Orbiter Mission reached Mars in

a) 2013

b) 2014

c) 2016

d) 2019

Answer: b

89.India became first Asian country to reach

a) Moon

b) Sun

c) Mars

d) Venus

Answer: c

90.PSLV-C37 launched how many satellites?

a) 20

b) 50

c) 75

d) 104

Answer: d

91.GSLV Mk III can carry

a) 1 ton

b) 2 tons

c) 4 tons

d) 6 tons

Answer: c

92.Chandrayaan-2 was launched in

a) 2017

b) 2018

c) 2019

d) 2020

Answer: c

93.Chandrayaan-3 landed on Moon in

a) 2019

b) 2020

c) 2022

d) 2023

Answer: d

94.Chandrayaan missions study

a) Mars

b) Venus

c) Moon

d) Sun

Answer: c

95.Big Bang theory explains

a) Planet motion

b) Star formation

c) Origin of Universe

d) Moon phases

Answer: c

96. Dark matter mainly affects

- a) Light
- b) Gravity
- c) Temperature
- d) Radiation

Answer: b

97. A galaxy may contain

- a) One star
- b) Few stars
- c) Billions of stars
- d) No stars

Answer: c

98. The Milky Way contains approximately

- a) 10 billion stars
- b) 50 billion stars
- c) 100–400 billion stars
- d) 1 trillion stars

Answer: c

99. Black holes are detected by

- a) Seeing them
- b) Their gravitational effects
- c) Sound waves
- d) Reflection

Answer: b

100. The Universe is still

- a) Contracting

b) Stable

c) Expanding

d) Collapsing

Answer: c